Perce-Neige Simulator



Driver's manual

Version 1.20.0

Drive the longest underground funicular in France: 3 474 m from Val Claret (2 111 m) to the Grande Motte glacier (3 032 m).

ARP273-ROSE • 2006 TI-Basic original • 2026 PyQt6 port • MIT licence

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1 Introduction

Welcome on board

This simulator faithfully reproduces the **Perce-Neige** underground funicular in Tignes, opened on 14 April 1993 by Von Roll (lead contractor) and CFD (rolling stock). It is the longest funicular in France (3 474 m), linking Val Claret (2 111 m) to the Grande Motte glacier (3 032 m) in under 7 minutes.

Who this manual is for

Even if you have never driven a train, a metro or a funicular, this manual teaches you to pilot the Perce-Neige step by step. Each phase of the trip, each button, each emergency situation is explained with the physical and technical *why* behind the real procedure. Data comes from the builder (Von Roll / CFD), public technical sources (remontees-mecaniques.net, Wikipedia) and French guided transport regulations (STRMTG).

What's new in October 2026 (v1.15.35 to v1.17.0)

The skier (`F9` on PC, SKIEUR button in the Web version): you play a skier who walks through the stations, boards the train, goes to the driver's seat and exits at the top, onto the terrace or through the Génépy piste door. The funicular runs by itself, waits for them and closes the doors once they are on board (section 14). **They ski**: skis on outdoors, down to Val Claret across a winter ski area, OpenStreetMap pistes groomed and marked, against the ghost of a user's real descents.
Driver's console reproduced from photos, with its live Pro-face screen; its buttons can be clicked, on PC as on a tablet (section 7).
Stations from the real thing: Val Claret as rebuilt in 2018 (waiting room, sliding doors to the platforms, pit under the track), Grande Motte open onto the glacier (terrace and its staircase, restaurant, cable car). **Real terrain** of the massif in 3D (IGN), "X-ray" exterior view, and on PC a true terrain cross-section in the side view.
Cable: catenary between the rollers, elasticity while running, rupture (it snaps, goes slack, the machinery stops). Faithful line rollers and numbered supports, emergency exit drawn from the video, each trip is 3 474 m on the counter.
Sounds: the horn is the real buzzer of the train; on the platform, gusts of air then silence.
Interface scaled to the screen, full screen (`F11`), overall volume (`F7`/`F8`), tunnel lighting (`J`), 3D tuned to the machine. **Linux** and **macOS** packages.

What's new in September 2026 (v1.15.0 to v1.15.34)

Audited physics: the rope's own weight (38 t, ±99 kN at the termini) and the tunnel air drag (piston effect) now enter the force balance; drive efficiency 0.90 / regeneration 0.80; roller resistance; ascent and descent finally give the same powers and tensions (mirror test). Details in `AUDIT_PHYSIQUE_VOYAGES.md`.
Doors timed on the sound: announcement, then buzzer (7 s), then the closing clip (7 s); the leaves start 1.3 s into the clip and hit the stop at 5.3 s. The interlock only switches at the stop.
Automatic update: the program installed by the Setup updates itself (ZIP archive, 3D viewer included); the Setup 1.15.3 or newer must be installed once.
3D view: realistic trains (windshield measured on photos, animated sliding doors, round halogen headlights with a warm-up, numbered plate), stepped floor, modelled passengers following the load with skis, poles and snowboards, smooth light tunnel lining, white rollers tilted in curves, rope at sleeper level, buried machinery at the top station (only the downhill sheave shows).

Welcome screen (F1) redesigned: three columns of shortcuts, driving tips underneath, nothing overlaps any more.

2 Installation

2.1 Simple use (recommended)

Everything is on the latest release page: https://github.com/ARP273-ROSE/perce-neige-sim/releases/latest

- **Windows**: PerceNeigeSimulator-Setup.exe. The installer goes into your user profile, no administrator password. On first launch Windows may show "Windows protected your PC" (unsigned installer): *More info*, then *Run anyway*.
- **macOS**: PerceNeigeSimulator-macos-arm64.dmg (Apple Silicon Macs: M1, M2, M3, M4...) or PerceNeigeSimulator-macos-x86_64.dmg (Intel Macs); drag it into Applications. The app is unsigned: on first launch, right-click → *Open*.
- **Linux**: PerceNeigeSimulator-linux-x86_64.tar.gz (or ...-linux-arm64.tar.gz). Unpack it and run ./installer.sh: the simulator installs into ~/.local/share, without sudo, with its entry in the applications menu.

On all three systems the program then updates itself (section 16). A former Linux AppImage (up to 1.19) moves to the new package by itself at its next update.

The 3D cabin view is included everywhere. No Python to install.

Web version (tablet, phone, browser): https://funiculaire.giff.re. It installs like an app ("Add to Home Screen"), works offline, and is driven with on-screen buttons (section 8).

2.2 From source (developers)

```
# Windows
launch.bat

# Linux / macOS
./launch.sh

# Manual install
pip install -r requirements.txt
python perce_neige_sim.py
```

3 What is a single-cable funicular?

3.1 The counterweight principle

A classical funicular is **not** a train that climbs on its own motor power: it is a *balanced* system. Both cars are tied to the two ends of the same cable, which runs over a **drive pulley** at the upper station. When one car goes up, the other *must* go down.

Key consequence

The weight of the descending car pulls downwards and **helps** the motor lift the ascending car. In theory, the motors only supply the *difference* in weight between the two cars, plus friction

losses and acceleration.

3.2 Why the descending car is (almost) always empty

On the Perce-Neige, skiers **go up** by funicular (packed: up to 334 passengers) and **ski back down** the glacier and pistes. The descending car is therefore almost empty all day (except during the summer glacier season).

- Ascending car, full: 32.3 t empty + 25.5 t load = **58.8 t**
- Descending car, empty: **32.3 t**
- Cable imbalance: up to **26.5 t**

These 26 tonnes of imbalance (plus acceleration and 30% grade) are what the motors must provide as *useful* force. Hence the **three 800 kW DC motors**, totalling **2 400 kW** (3 260 hp), installed at the upper station.

3.3 The passing loop: Abt system

The tunnel is **single-track** over almost its full length (cheaper to excavate), but both cars must cross somewhere! At mid-slope, over 203 m, the tunnel splits into **two separate bores** — the bypass tube was holed through in October 1991, one day after the main bore — following the **passing loop** principle patented by Roman Abt in 1879. There are **no moving switches**:

- Each car's bogies are **asymmetric**: on one side, double-flanged wheels (guided, imposing the trajectory); on the other, plain cylindrical wheels (free to cross the frog and the opposing cable).
- The two cars have *opposite* asymmetry, so they **naturally** take one the right track, the other the left.
- They always cross at the same point, at the same speed, in opposite directions.

In the 3D view the switch is drawn like the real one. The outer rails are continuous. Each inner rail starts with a *tongue* born in the single track, next to the opposite outer rail: the other car's flange runs between them. The rail is then cut where the opposing car's cable crosses it. There the two rail ends are bent to run side by side, parallel to the cable, 8 cm on either side of it, over 1 m; they sit on bolted plates, and the cable slides between them on a steel plate. The 24 cm plain wheel rides on one, then the other. The two inner rails finally cross in an X on a cast frog. Inclined deflection sheaves guide the cable through the bend. Everywhere, the cable runs in a straight line from one roller to the next: in curves it passes up to 5 cm inside the track between rollers.

In the simulator, the mini-map at the top shows both cars converging; view F4 shows the tunnel *forking into two bores* (unlike its Val d'Isère sibling, the crossing here happens in two separate tubes, each car in its own). No need to slow down: crossing happens at cruise speed.

4 The Von Roll regulator: setpoint, not throttle

4.1 Principle

On the Perce-Neige, the driver **does not pilot** like a motorist. There is no throttle that directly feeds motor power. Instead, the driver sets a **speed command** (setpoint), as a percentage of maximum speed.

Regulation loop

The Von Roll electronics continuously compare:

1. The **actual speed** measured by an encoder on the drive pulley,
2. The **command** set by the driver.

It then adjusts the DC motors' field current to track actual speed to the command, within the authorised acceleration envelope (∼1 m/s^2 max for passenger comfort).

4.2 Why not a direct throttle?

- **Guaranteed profile**: the car always accelerates the same way, regardless of weight, local grade or driver mood. Passengers are never surprised.
- **Automatic limitation**: impossible to exceed 12 m/s, even at full command. Above 110% of that speed, the safety brake triggers automatically.
- **Programmed station approach**: when the car nears a station, the command is automatically capped to ensure controlled deceleration and precise platform stop.
- **Reproducibility**: every trip is identical. Comfort, energy and schedule can be guaranteed.

4.3 What the driver still controls

The driver retains:

- The **command** (0–100%, by steps).
- The **departure order** (doors closed, bell).
- The **electric stop** (latched button).
- The **emergency stop** (red mushroom button).
- The **dead-man vigilance**.
- The **accessories**: headlights, cabin lights, horn, doors, announcements.

This is **supervised driving**, as opposed to manual train driving.

5 A complete trip, phase by phase (Normal mode)

Let's imagine a Val Claret → glacier ride. Here is what happens from start to finish, and what you must do at each step.

5.1 Phase 0: pre-departure checks

Real pre-departure checklist

Before the first ride of the day, the real Perce-Neige driver verifies:

- Dead-man test (hold the pedal >60 s: an alarm must sound at ∼58 s, then automatic stop 3 s later).
- Brake test: service brake then safety brake, empty, over a few metres.
- Cable tension reading (∼22 500 daN nominal).

- Hydraulic levels, motor temperatures, 400 V auxiliaries.
- Radio check with the opposite station.
- Platform lights and departure bell test.
- Door condition (open/close test).
- Upper station weather (temperature, icing, wind: at 3 032 m, −20 °C is routine).

In the simulator these checks are implicit. You arrive in the cabin with the train ready to go.

5.2 Phase 1: closing doors and arming

Driver actions

1. Key `D`: close doors. A green "doors closed" light comes on.
2. Departure impossible until doors are locked: this is a hardwired **safety interlock**, you cannot bypass it.

Why?

Funicular doors open directly onto the void (platform-level door, circular tunnel). Opening them in motion would be lethal: hence the absolute lock until the car is fully stopped at a platform and aligned.

5.3 Phase 2: departure, the gentle ramp

Driver actions

Hold the up arrow (↑): the speed command rises continuously (35% per second) while the key is held, the down arrow lowers it. There are no steps, and `W` is not a command key (it toggles the vigilance, see below). For a comfortable start, leave the command at 100%: the regulator does the gentle ramp itself.

Mechanically: the regulator commands the motors to progressively overcome static force (dry friction, gravity on the grade). The car eases into motion with a controlled **acceleration ramp** of about 0.3–0.5 m/s^2, or 1–2 km/h per second. Very gentle: standing passengers are not unbalanced.

Physics note

The grade at the start is **18%** (relatively gentle). On this section, required motor torque is minimal because the counterweight (descending car nearing the upper station) already pulls upward.

5.4 Phase 3: speed build-up, dead-man vigilance

Driver actions

Gradually raise the command to 100% (12 m/s). If the vigilance is on (`W`, **PC only**: the Web version has no vigilance), **at least every 20 seconds**, touch a control (arrow key, headlight, horn, or `G` "acknowledge"). Otherwise, automatic stop.

Dead-man vigilance, in detail

The **automatic vigilance device** (colloquially "dead-man's pedal") is required by regulation for all guided transports: trains, metros, trams, funiculars. Its purpose: if the driver loses

consciousness, falls asleep or is absent, the train *must* stop automatically.
On the Perce-Neige:

- The driver must act (touch pedal or button) at least once every 20 seconds.
- A *held* press is not enough: you must **release and re-press**. You cannot "block" the pedal with a rock. This is the fundamental principle.
- If the delay is exceeded, a sound alarm rings. Without action within 3 further seconds, **automatic electric stop**.

In the simulator, any driver action (arrow, key, etc.) resets the timer. Key `G` acknowledges without changing anything else.

5.5 Phase 4: cruise at 12 m/s

Driver actions

None: the regulator holds 12 m/s alone. You monitor instruments, rearm vigilance every 15–20 s.

What you watch:

- **Analog speedometer** in m/s (green up to 11, yellow 11–12, red beyond).
- **Tension gauge** in decanewtons (daN): shows the *most loaded rope side* at the drive pulley — each side carries its own car weight along *its* local grade plus the hanging cable self-weight (11 kg/m: up to ∼9 900 daN with a car at the bottom). Service scale 0–42 000 daN: green up to the 28 000 alert (nominal 22 500 is a service marker, not an alarm), orange 28–35 000, red beyond; rupture (191 200 daN, safety factor ∼8.5) is off-scale. A *sharp drop* = cable slack = serious incident. A *rise* = abnormal load or friction.
- **Motor power gauge**: proportional to grade and load. A spike = brake dragging or obstacle.
- **Mini-map at top**: both cars advance cell by cell; you see the opposite car descending towards you.

5.6 Phase 5: crossing at the passing loop

Mid-way, you enter the tunnel doubling. The opposing car passes on your left (climbing) or right (descending) at 12 m/s. No action needed: asymmetric bogies handle everything. The simulator shows the second tunnel opposite in F4 view.

Relative speed

12 m/s + 12 m/s = 24 m/s relative, or 86 km/h! Crossing takes about 8 seconds over the 203 m loop.

5.7 Phase 6: station approach

What the regulator does automatically

When the car passes a marker at ∼120 m from the platform, the command is automatically capped. The regulator applies a **deceleration ramp** at ∼0.3 m/s^2 (electric braking: motors act as generators and absorb kinetic energy), bringing speed from 12 m/s down to 1 m/s over the last 100 metres before the platform.

You see the speedometer drop by itself. You may add some service brake (`Space` or `B`) to emphasise, but it is not needed.

5.8 Phase 7: the creep zone

On the last 55 m before the platform, the car rolls at **0.75 m/s** (2.7 km/h, real-world measured value). This is the **creep zone**. Over the final 2.5 m the regulator tracks a $\sqrt{2ad}$ docking taper (0.15 m/s^2): speed melts continuously — 0.4, 0.2, 0.08 m/s over the last three seconds — down to a jolt-free stop.

Why so slow?

- **Alignment precision**: platform doors and car doors must match within centimetres.
- **Passenger safety**: stopping from 12 m/s takes ∼30 m; from 1 m/s, 10 cm is enough for any hiccup.
- **Clearance**: the platform has protruding equipment (cameras, sensors, anti-fall devices) that the car must approach without touching.

5.9 Phase 8: stop and door unlocking

Automatic sequence

1. The car stops exactly at the alignment marker.
2. The regulator commands the parking brake.
3. Motors are de-excited.
4. The system detects "immobilisation confirmed".
5. Platform doors *and* car doors open in sync.

In the simulator: key `D` to open (once stopped). The "Tignes, doors opening" announcement plays automatically.

5.10 Phase 9: cable elasticity — rebound and boarding sag

Two very specific Perce-Neige details, both emerging from the same physics (rope-side stiffness $k = EA/L$, L = cable length from the car to the top drive pulley):

The arrival rebound

When a car stops at the *lower* station its rope side is ∼3.45 km long: stiffness is low and the energy released as the drum grabs excites a visible damped oscillation (±25–45 cm, ∼7 s period). At the top station the side is only ∼20 m: millimetric, invisible. The asymmetry falls out of the maths — nothing is hand-tuned.

The boarding sag

At the lower platform, every boarding passenger stretches the rope elastically ($\delta = \Delta m\, g \sin\theta\, L/EA$, ∼1.6 mm per passenger): the car **slowly sinks by up to ∼40 cm** while loading, and stays at the sagged position until departure — the elongation persists as long as the load is there. The offset quietly dissolves once under way. At the top station the effect is millimetric.

6 Each button in detail

6.1 Driving controls

Key	Role
↑	Raises the speed command while held (in % of 12 m/s). Available unless the car is under electric or emergency stop. This is your primary control.
↓	Lowers the command while held. The car brakes electrically (generator mode) down to the new command. Use: reduce speed near an incident, or for intermediate cruising.
`Space / B`	**Service brake** (held): mechanically bites on the drive pulley. Deceleration ∼2.5 m/s^2. Useful in addition to electric braking, or for shorter stops. Release the key, brake releases.
`Shift`	**Commanded emergency stop**: safety brake on the drive pulley, ≤1.25 m/s^2 (standing-passenger limit; the 3.6 m/s^2 parachute only drops on +20 % overspeed or cable rupture). Use only in emergencies (visible obstacle, signal violation, immediate anomaly). Repeated use = shoe wear.
`3`	**Latched electric stop**: engages a permanent electric stop (car decelerates and stays stopped) even if you release. Press `3` again to resume. Equivalent to a "held neutral" for minor incidents.
`4`	**Latched emergency stop**: the *same* safety brake on the drive pulley as `Shift` (1.25 m/s^2), but latched — it stays applied when you release the key, and the drum re-engages at standstill. Used for any serious incident: cable fault, smoke, intrusion. Press `4` again to release, car stopped (simulator only: in reality, control centre intervention required). The rail clamps (parachute, 3.6 m/s^2) are not a command: they drop by themselves on +20 % overspeed or cable rupture.
`V`	**READY**: car stopped, *doors closed* (`D` first); the driver of the other car confirms 2 to 4 s later (PRÊT VÉHICULE light). After an arrival, the first `V` does the silent turnaround (direction reversal), the next one arms READY. Mid-tunnel, READY plays "Resuming service". Refused with doors open (except in Challenge), under electric stop or with a fault active.
`Z`	**forced START**. In service there is no START button any more: READY armed, the other cabin ready (2–4 s), doors closed and no traction interlock (emergency, electric stop, vigilance, fault, setpoint at 0), the start follows **by itself** — platform buzzer (8 s at the bottom, 6 s at the top), then traction, like the Web version. The normal sequence is therefore `D` (closing: announcement, buzzer, clip) → `V`, and that is all. `Z` forces the start without waiting; only Challenge mode lets you leave with the doors open (`Z` does not close them: reckless start).
`I`	**Reverse direction**, train stopped. At a station it is the normal, silent turnaround; in the tunnel, the "return to the station" announcement plays.
`W`	**Vigilance** (dead-man) on or off. Optional, **PC only** (not in the Web version); once on, touch a control every 20 s or the train stops by itself.

Key	Role
G	**Vigilance acknowledge** (dead-man): resets the 20 s timer. Equivalent to pressing the vigilance pedal. No other action, but lets you restart the timer without touching other controls.

6.2 Cockpit controls

Key	Role
H	**Headlights** on/off. Off by default. Turn on to see further in the tunnel (F4 view): the beam reaches ∼100 m with exponential falloff. In reality, the car's lights illuminate 80–120 m ahead.
C	**Cabin lighting**: bright or dimmed. On the real car, ceiling fluorescents can be dimmed for visual comfort during long stops.
J	**Tunnel lighting** on or off. Off, the tunnel is pitch dark; only the train's headlights light it.
K	**Horn** (held): external audible warning. Used on the real car to signal a departure, a hazard, or to greet. The sound is the train's real buzzer.
D	**Doors** open/close, **only when fully stopped** at a platform. Absolute lock during the trip.
A	**Trip autopilot**: the simulator drives one full trip by itself — closes the doors, arms READY, gives START, holds 100 % setpoint, lets the stop envelope land the train — then disengages on arrival. Shows the exemplary procedure. It hands back control as soon as you touch the setpoint or a brake, on a fault, or when auto-operation (X) takes the line; engaged after an arrival, it reverses and departs again.
X	**Auto-exploitation**: ambient mode. The simulator runs the full published service unattended from opening (08:45) to last descent (16:45) — boarding, doors close, trip, crossing, turnaround — at peak cruise (12 m/s) and off-peak cruise (10.3 m/s) according to the real clock. Every round-trip, passenger load and incident is logged to `exploitation.db`.
Shift+X	**24/7**: auto-operation ignores the published hours.
N	**Sound**: mutes or restores all sound (ambience, announcements, buzzers). The recordings are authentic.
Backspace	Aborts the current announcement.

6.3 System controls

Key	Role
P	Pause / resume simulator.
M	Mode cycle: Normal → Challenge → Faults. See next section.
L	Toggle FR ↔ EN interface.
F1	Welcome / help screen: three columns of shortcuts (Driving, Cockpit, System) and driving tips underneath; shown at launch, F1 closes it.
F2	Manual announcement console (trigger any message on demand).
F3	Real machine info (specs, sources).
F4	First-person cabin view (realistic pinhole). At launch, when the 3D viewer is available, the cabin view is directly the embedded Godot 3D; F4 keeps its cycle (off, drawn, 3D).
0	3D views in turn: cabin, exterior orbit, then the **machine room**. Drag to orbit (left mouse held / one finger), wheel to zoom (two-finger pinch on touch). In the machine room the camera does not follow the train: it turns all around the two drive wheels, right drag (or two sliding fingers) moves it; it slides freely from the room floor up to the hall ceiling, through the station slab, and walls in the way fade out (section 7).
F9	**Skier** in the 3D view (section 14); E puts the skis on (section 14.9).
F5	Auto-operation trip log.
F6	Open the documentation download dialog (manual + theory guide PDFs from GitHub).
F	In Faults mode only: open the fault picker dialog (15 faults grid + auto/manual scheduler toggle).
+ / - / 0	Zoom the side-profile view (or mouse wheel). Aspect ratio is locked: the visual slope angle stays at the real slope angle at any zoom level.
R	New trip (after arrival) ; at a platform, stopped : maintenance acknowledgement of a non-catastrophic fault.
Enter	Start from title screen.
Home	Back to the title screen (choice of train and direction).
Esc	Pause / menu / quit.
F7 / F8	Overall volume − / + (also: mouse wheel on the gauge next to SOUND).
F11	Full screen / window.
Help menu	Shortcuts (F1), manual and guide as PDF (F6), the real machine (F3), skier (F9), what's new (online version history), update, report a problem, automatic incident reports, about.
Display menu	3D quality (automatic or forced), interface size, full screen.

7 The 3D views and the driver's console

7.1 3D cabin view

F4 (or the 3D VIEW button) shows the embedded 3D view: nothing to install. The trains are drawn from photos (double-flanged Abt wheels, round halogen headlights, numbered plate); inside, a black rubber landing level with each window, blue benches along the walls and orange ski racks; the passengers are skiers with skis, poles and snowboards, according to the load.

Shuttle: the train never turns round, it sets off the other way. It has a driving desk at each end (desk, seat, clock, monitor); both are powered and always show the readings. You drive from the leading desk: the front (car 1) going up, the rear (car 2) going down — the cab view, the headlights and CONDUIRE (skier) move there by themselves at departure.

Magnifier: at 1080p the Pro-face screen is only 230 pixels wide and its figures cannot be read. The mouse wheel (or a pinch, or the LOUPE button on touch screens) turns the camera towards the screen and narrows the field of view from 78° to 32° in three steps; everything becomes legible, the enlarged buttons can be clicked, and a wheel step back restores the normal view.

7.2 The console

In front of the driver, the console is reproduced from photos of the cab: on the left the ARRÊTS (stops) frame, in the middle the Pro-face touch screen, on the right the button plate. Its controls can be clicked (or touched on a tablet). On PC, each button presses the key that does the same thing, with the same interlocks:

Control	Role	Key
PORTES 1 À 6 / 7 À 12: OUVERTURE (white), FERMETURE (green)	Open or close the doors; 1 to 6 on the left looking uphill, 7 to 12 on the right. The lamps follow the doors.	D
PRÊT	Green lamp: the train is ready to depart.	—
MONTÉE	"We are ready": READY, then on the second press START.	V, Z
− VITE / + VITE	Spring-return selector: held left or right, it lowers or raises the setpoint.	↓ ↑
EN MARCHE (key)	Main switch: when off, the console goes dark and MONTÉE is refused.	—
KLAXON	While pressed.	K
ÉCLAIRAGE: CABINE (0/1), COMPARTIMENT	Cabin lighting; SECOURS (emergency lighting) is always on.	C
ARRÊTS: URGENCE (big mushroom)	Emergency stop, latched down; press again to release.	4
ARRÊTS: ÉLECTRIQUE (small)	Electric stop, latched the same way.	3
Pro-face screen	Local date and time, speed to the hundredth, distance on the counter (0 at departure, 3 474 m on arrival), train lamps; a touch switches the page (driving / doors).	—

7.3 Exterior view

`O` switches to the orbiting exterior view. The massif is in 3D, over 21 × 17 km: the whole Tignes – Val d'Isère area, from Les Brévières to the Pissaillas glacier via Le Fornet (IGN terrain and orthophotography). Around the tunnel the ground turns translucent: the whole line shows through the mountain, and you can pass under the track. Drag to turn, wheel (or pinch) to zoom.

7.4 Machine room

A second `O` leads into the machine room of the top station: the two yellow drive wheels, the cable in a figure eight, the real sound of the station recorded at 12 m/s. The camera is free; right drag moves it.

7.5 Side view (PC)

Without the cabin view, the PC shows a true cross-section of the terrain along the line, taken from the IGN terrain, with the trains and stations. Zoom: `+` / − / `0` or the wheel.

7.6 3D quality

At launch, the 3D reads the machine (processor, memory, graphics card, screen) and picks its settings; if the frame rate drops, it removes effects one by one, in this order: indirect lighting, fog and reflections, anti-aliasing, glow, then a 30 fps lock, and only as a last resort 85 % rendering — never less on PC, so that the console's buttons and screen stay legible. Each change is written to the log ("3D view: graphics level 3/6 — no anti-aliasing"). The **Display → 3D quality** menu forces high, medium or low.

8 The Web version (tablet)

On a tablet, everything goes through the on-screen buttons:

Button	Role
+ VITESSE / − VITESSE	Setpoint, while pressed.
FREIN	Service brake, while pressed.
URGENCE	Emergency stop.
PORTES	Open or close, train stopped at a station.
PRÊT / DÉPART	One-button departure sequence: announcement, doors closing, other car's confirmation (2–4 s), buzzer, traction. Refused (message) with the command at 0, the emergency applied, an uncycled door fault or a serious fault; a second press releases an emergency.
INVERSER	Reverse direction, when stopped.
AUTO	Auto-operation: 12 m/s at peak hours (9–12 h, 14–16 h), 10.3 m/s otherwise, passenger load by time of day.
PHARES, CABINE, TUNNEL	Lighting.
VUE	Cabin, exterior, machine room.
SKIEUR	The skier (section 14).
ANNONCES	Audio announcement menu; STOP cuts the announcement, FERMER closes the menu.

Button	Role
MODE, PANNE	Game mode; fault choice in Faults mode.

With a keyboard (computer browser): ↑/↓ setpoint, `Space` brake, `Shift` emergency, `Enter` start, `D` doors, `I` reverse, `H` headlights, `C` cabin, `J` tunnel, `V` or `O` view, `M` mode, `F` fault choice, `R` new trip after an accident or a serious fault, `F3` auto-operation, `K` skier (then `E` for the skis).

Differences from the PC

- **Arrival**: the Web version opens the doors and reverses the direction *by itself*, once the cable has settled (3 to 45 s after the stop depending on the oscillation), even in manual driving. On PC, `D` opens and the first `V` reverses.
- **Vigilance** (dead-man, `W`/`G`): PC only.
- **Stop in the tunnel**: the Web version has no abnormal-stop sequence (announcement at standstill, READY "Resuming service"); after an emergency stop, PRÊT / DÉPART releases the emergency and the command restarts the car.

9 Order of operations and restrictions

9.1 Authorised sequence

The simulator replicates real safety interlocks. Some actions are **impossible** depending on the car state:

- **Doors**: opening only with the car nearly still (speed < 0.2 m/s) and at a station (no interlock in Challenge).
- **Command**: always adjustable, even at the platform or under a stop. The *traction* interlocks are what count: no torque without the READY/START sequence, with doors open (outside Challenge), under electric or emergency stop — and START refuses a command at 0.
- **Emergency stop**: always overrides everything.
- **Emergency release**: impossible while a *cause* is active (cable, thermal, door, etc.). The incident must first resolve or be handled.

9.2 What you cannot do

- Exceed 12 m/s: the regulator clips (100% command = 12 m/s). Except in Challenge mode, where the command goes up to 15 m/s and every safety net is gone.
- Open a door in motion (outside Challenge).
- Depart at full speed instantly: the acceleration ramp is capped at 0.32 m/s^2 (0.12 when pulling away from the platform), whatever you request.
- Force a door to close on an obstacle (foot-sensor in reality).

10 Emergency situations

10.1 Electric stop vs emergency stop vs rail brake

This is the most important distinction for a funicular driver:

Type	Mechanism	When?
Electric stop	DC motors switch to **generator mode**. Torque reverses, car decelerates naturally. No mechanical brake bites. Gentle 0.3–0.8 m/s^2.	Normal braking at station approach. First step for minor anomaly (door sensor, light thermal warning).
Emergency stop (safety brake)	**Mechanical shoes** biting on the drive pulley at the upper station. Deceleration up to ~1.25 m/s^2. Wear at every use.	Mushroom button. Dead-man not acknowledged. Overspeed >110%. Critical fault (cable, fire).
Rail brake (parachute)	**Clamps on bogies** closing on the rail head. Deceleration ~3.6 m/s^2 (3.2–4.1 measured on Belleville clamps; absolute STRMTG ceiling 5 m/s^2) (passenger-comfort limit). Stopping distance 8–35 m depending on grade and direction (see § Cable rupture calc). Passive spring mechanism, no electronics.	Cable rupture (slack detection). Extreme overspeed. **Never used in normal operation.** Last line of defence.

Remember

The **rail brake is passive**: it depends on no electricity, no hydraulics, no electronics. If everything fails, it drops by itself. That is why a funicular is statistically safer than a lift.

10.2 Emergency scenarios and expected reactions

10.2.1 Cable tension anomaly

Tension gauge leaves the green band.

- Moderate rise: slow down (lower command).
- Sharp rise ($\geq$ 150% nominal): electric stop, diagnose.
- Severe drop (cable slack): immediate emergency stop, rail brake automatically armed.

10.2.2 Motor thermal overload

One of the three motors overheats.

- The regulator automatically reduces command.
- In the simulator: a red "FAULT" light, logged event.
- Drive at reduced speed to the nearest station; do *not* stop in the tunnel unless forced.

10.2.3 Door fault in tunnel

A door sensor fault mid-ride.

- **Never open** a door in the tunnel.
- Electric stop, passenger announcement.
- Resume gently to the nearest station, then manual platform inspection.

10.2.4 Smoke / fire

Smoke detector triggers.

- Immediate emergency stop (`4`).
- In reality: tunnel ventilation reverses to push smoke away from the passenger evacuation direction.
- Evacuation via the marked emergency walkway (side passageways).

10.2.5 Icing at the upper station

At 3 032 m, platform doors can freeze.

- Reduce command to 6–8 m/s for final approach.
- Extended creep zone.
- Door de-icing cycle (simulated).

10.2.6 Dead-man not acknowledged (PC only, vigilance enabled with `W`)

No control touched for 20 s.

- Sound alarm at 18 s.
- At 20–21 s, automatic electric stop.
- Resume after acknowledge (`G`).

10.2.7 Overspeed

Speed $> 110\%$ of `V_MAX` $=>$ 13.2 m/s.

- Automatic emergency stop, safety brake.
- Impossible in normal driving (regulator caps), but can happen in Faults mode (descent, runaway cable).

10.2.8 Cable rupture (theoretical)

Catastrophic scenario, never observed on the Perce-Neige. The 8.5 design safety factor (191 200 / 22 500 daN) makes the event almost impossible, but the system is engineered to handle it.

- Cable slack detected in < 0.1 s by differential tension sensors.
- Rail brake (Belleville-spring parachute) drops automatically, jaws fully closed in 0.3–0.5 s.
- **Progressive** stop, not instant : distance depends on mass, local grade and direction (see calculation below).
- Brake deceleration $\sim$3.6 m/s^2 (measured practice of Belleville clamps; the absolute STRMTG ceiling — 5 m/s^2 for standing passengers — is not the operating value).
- Directed evacuation by control centre via lateral walkways once immobilised.

Stopping distance calculation

Kinetic energy to dissipate : $E_c = \frac{1}{2}mv^2$. On a descent, gravity *adds* energy during braking. Net deceleration on grade θ :

$$a_{\text{net}} = a_{\text{brake}} - g \sin\theta$$

and stopping distance :

$$d_{\text{stop}} = \frac{v^2}{2\, a_{\text{net}}}$$

The steeper the grade, the longer the distance (gravity fights the brake).

Numerical examples (rail brake at 3.6 m/s^2 — Belleville practice —, rupture speed 12 m/s)

Scenario	Grade	a_{net}	Distance
Full rame 58.8 t, descent	30 %	0.78 m/s^2	$\approx$ 92 m (15.4 s)
Full rame 58.8 t, descent	18 %	1.86 m/s^2	$\approx$ 39 m (6.5 s)
Empty rame 32.3 t, descent	30 %	0.78 m/s^2	$\approx$ 92 m (15.4 s)
Full rame, ascent	30 %	6.42 m/s^2	$\approx$ 11 m (1.9 s)

Fundamental asymmetry

On **ascent**, gravity helps the brake : short stop (8–12 m). On **descent**, gravity fights the brake : long stop (25–35 m). When the cable is intact, it links both rames : the ascending one brakes the descending one through the pulley. That is an essential passive safety. Upon rupture, this link is lost and each rame must stop on its own parachute.

11 Why these technical choices?

11.1 Why 1 200 mm track gauge?

Compromise between tunnel clearance and stability:

- Wider than metre gauge (1 000 mm): stability for a 31.6 m $\times$ 3.55 m car on 30% grade.
- Narrower than standard gauge (1 435 mm): minimal tunnel cross-section. Diameter 3.9 m instead of 4.5 m, i.e. $\sim$30% less rock to excavate over 3 474 m. Major saving.

11.2 Why three 800 kW motors?

- **Triple redundancy**: degraded service possible on 2 motors (67% torque = reduced speed but service maintained).
- **DC**: superior speed regulation vs asynchronous motors of that era (1989–1993), high torque at low speed.
- **2 400 kW (∼3 260 hp)**: needed to accelerate 26 t of imbalance at 0.5 m/s^2 against 30% grade and cumulative friction.

11.3 Why a 52 mm cable?

- **Safety factor ≥ 6** required by STRMTG regulations. Theoretical breaking load ∼135 000 daN, working tension 22 500 daN.
- **Compatible with ⌀ 4 160 mm drive pulley**: D/d ratio 80, ideal for bending fatigue (cable longevity).
- Heavy-duty funicular standard: steel core, multi-strand, shock and abrasion resistant.

12 Game modes

12.1 Normal mode

A simple Val Claret → glacier then return. No incident, perfect conditions. Ideal for learning.

12.2 Challenge mode

Manual driving, scored out of **100** at arrival — you handle everything: *the Von Roll auto-docking safety net is removed.*

Scoring (press `M` to enable)

- **Comfort** (40 %): smooth driving. Any hard accel / decel (beyond ∼0.9 m/s^2 standing), jerk, and above all any emergency stop tanks the score (ISO 2631 model).
- **Stop precision** (35 %): stop *right* on the platform mark (100 at ≤0.10 m, 0 at ≥2 m).
- **Discipline** (25 %): no emergency stop (−60) nor fault (−40).

A result banner (score /100 + stars) shows at arrival, the stop mark is shown live on the control panel, and the **best score is saved**.

Danger: no automatic safety The controller no longer brakes for you. **Cut the setpoint in time** (service braking ∼1.2 m/s^2: ∼60 m to stop from 12 m/s). The red **"TOO FAST"** alarm lights as soon as your speed exceeds the profile the automation would hold here — purely advisory, it does not brake.

Runaway and overspeed Leave the **setpoint at full** on a loaded descent and the speed cap is lifted: gravity **runs the car away** past 12 m/s. Brake yourself (**Space** = service brake, **Shift** = emergency). Otherwise, at **+20%** overspeed the motor control lets go: **motor destroyed, cable snapped** (you get a message). With the strand **severed**, the **tension gauge drops to zero**. Decoupled from the counterweight, the car **free-falls** down the slope — only the parachute brake (Shift) can still stop it before disaster. Beware: on a ∼30 % grade the parachute (3.6 m/s^2) fights gravity (∼2.8 m/s^2), a *net* braking of only ∼0.8 m/s^2 — **apply it very early**; stopping distance grows with the square of speed (∼550 m from 30 m/s). If the rope snaps *on the way up*, the car coasts on, stops for an instant at the top of its travel, then **rolls back down** and picks up speed: neither the top-station drum nor the incident sequence can hold it, the rope being cut.

Reckless start: every interlock gone In chaos mode you can **leave before the other car is ready**, **doors open**, mid-boarding. Every breach is scored (*RECKLESS START*, −50 discipline) and triggers sarcastic lines. You can **drive doors open** for the whole trip (no auto-close; `D` any time) and do a **U-turn**: each earns its jabs, and the passenger reviews reflect it.

Motor over-rev & zero setpoint In Challenge the electronic limiter is gone: **full setpoint** lets the motor over-rev past V_MAX *even on a loaded climb*, up to the overspeed cascade. The setpoint is still a **speed setpoint**, zero included: reducing it brakes through the drive (0.7 m/s² in Challenge, plan ahead), and at **setpoint 0** the drive stops the car and **holds** it. There is no free-wheeling at 0 any more: with no tail rope, the counterweight's 3.4 km of rope pulls about 99 kN towards the top station at the end of a climb, and a car released at setpoint 0 used to re-accelerate into the buffer. Whenever a brake is applied, **traction is cut** — the motor never pulls against the brake.

After a rupture: the parachute is the only recourse Cable severed, the pulley brake has no purchase: **only the Belleville rail parachute acts**. The emergency (**Shift**) engages it directly and it genuinely brakes (3.6 m/s², net ∼0.8 on the grade). There is **no automatic stop** in Challenge — you brake.

The possible endings

- **Buffer collision**: arrive too fast at the mark and the car rolls to the *actual* wall and hits it (screen shake, cable jolt).
- **Derailment**: taking the central Abt passing loop in overspeed (> 13.5 m/s, above certification tolerance) throws the car off the rails.
- **Head-on**: cable snapped, the other car has applied its brake and stopped where it was at the moment of the rupture (not necessarily at the passing loop) — yours, running away, plows into it.

Each game-over shows a sarcastic line and a 1-star **passenger review** styled by nationality — the apologetic Japanese, the German who timed the deceleration, the Italian "Mamma mia!"... A **flawless** run earns equally typed 5-star reviews (original language *and* English translation). The bank holds about a hundred quips and nearly a hundred reviews (twenty nationalities); at the buffer, the quip matches the impact speed, and a recently drawn line is not repeated. In skier mode, a few quips (at most one every 40 seconds, right when it happens) greet long off-piste runs, a missed funicular or an evacuation. Others greet a reckless start with the doors open, doors opened mid-line, overspeed and a too-fast station arrival. Press **R** to retry (also in skier mode): after a crash on arrival, the new trip departs **from the station you crashed at** (the return leg), not from the original start.

12.3 Faults mode

Fifteen incidents calibrated on real documented funicular failures (STRMTG, BEA-TT, Kaprun 2000, Glória Lisbon 2025, Carmelit, Perce-Neige 2008, Montmartre 2006, Sassi-Superga, M2 Lausanne) are available. By default a weighted random scheduler picks one; press `F` to open the *Fault picker* dialog where you can either trigger a specific fault now, or turn off the auto-scheduler to play manually.

Common operational faults `tension`, `door`, `thermal`, `fire`, `wet_rail`, `motor_degraded` (M1/M2/M3 named, Sassi-Superga precedent), `slack`, `aux_power` (Perce-Neige 2008 pattern), `parking_stuck`.

Severe / catastrophic `cable_rupture` (Glória Lisbon class), `service_brake_fail` (Glória double-failure), `flood_tunnel`, `comms_loss` (Kaprun lesson), `switch_abt_fault`, `fire_vent_fail` (fire + smoke vent down, Kaprun class).

Recover to the nearest station Faults that **halt the car in the line** (400 V lost, parking brake stuck, switch misalignment) require, once stopped, limping to the **NEAREST station at reduced speed** (≤ 3 m/s) — possibly **reversing direction** (`I`) if that station is behind. The log names the station, the distance and whether to turn back. The fault clears on arrival (maintenance at the platform).

Handling under automatic operation In **Faults + auto-operation** (`X`) the AI driver **lets incidents happen and handles them alone**: it limps to the nearest station (reversing if needed) for blocking faults, and after a catastrophic fault waits out the evacuation then **restarts service automatically (R)**.

Three-stage overspeed cascade (EP0392938A1 / STRMTG RM5) +10 % service brake, +12 % secondary emergency, +20 % mechanical Belleville centrifugal parachute trip.

Cable cumulative fatigue Per-trip `fatigue_cycles` counter plus time-integrated `cable_wear_pct` (Palmgren-Miner / ISO 4309 / DIN EN 12927-6). Real Perce-Neige cable replaced in 1999 after 6 years (≈36 000 round trips).

Realistic recovery (v1.9.0) Faults are now classified by *severity* and each category has its own recovery path. A persistent on-screen panel (red for catastrophic, amber otherwise) tells the driver what is happening, what to do, and what is blocked. Since v1.12.20 it lives in the **bottom strip**, between the event log and the auto-ops panel — the 3D view stays visible during the fault.

- **Advisory** (`tension`, `wet_rail`, `slack`, `comms_loss`) — dashboard warning only, no operational impact, auto-clears on the timer.
- **Operational** (`door`, `thermal`, `motor_degraded`, `flood_tunnel`) — degraded mode (speed cap, power derate), trip continues, auto-clears on the timer.
- **Stopping** (`aux_power`, `parking_stuck`, `switch_abt_fault`) — the cabin must stop ; *releasing the brake alone is not enough to resume*, you must press **READY (V) + DEPART (Z)** like at boarding.
- **Catastrophic** (`cable_rupture`, `fire`, `fire_vent_fail`, `service_brake_fail`) — trip is **TERMINATED**. The state machine runs `active` → `intervention_called` (*tech incident* PA) → `evacuating` (*dim_light + evacuation* PA, cabin lights dimmed, passengers evacuated) → `out_of_service`. READY and DEPART are refused indefinitely with the message *"trip terminated by <fault> — press R for new trip"*. The only way out is to press **R** for a new trip from the menu (real-world equivalent: maintenance call-out, on-site intervention, return to service after inspection).

Automatic safety chains (v1.12.21) The faults physics audit gave the monitoring system the same automatisms as the real installation — no serious fault lets the cabin keep rolling any more:

- **service-brake pressure switch**: brake fade while moving → automatic emergency stop in ∼3 s;
- **progressive fault speed ceiling**: a cap (wet rails, thermal, flood…) is reached through a 0.60 m/s^2 setpoint ramp — no more instant motor cut; if speed stays > cap + 1 m/s for 12 s *without firm deceleration*, the monitoring trips the emergency;
- **slack-cable switch**: deceleration > 1.5 m/s^2 during a slack fault → emergency;
- **tension red threshold** (35 000 daN) during a tension surge → emergency;
- **power-loss brake**: losing the 400 V auxiliaries clamps the spring-applied safety brake instead of leaving the cabin coasting; a stuck drum brake while moving actually brakes.

The electric stop now follows the Von Roll doctrine: setpoint capped at the current speed then a 0.45 m/s^2 regenerative ramp, ≈0.4 m/s^2 deceleration in both directions (REGEN gauge active).

Maintenance acknowledgement (v1.12.22) A *non-catastrophic* fault with the cabin **at a platform and stopped** can be lifted immediately with **R** (real-world equivalent: the technician fixes it on the spot): fault cleared, emergency released, overspeed latches reset, departure possible without waiting out the timer (35–90 s). Mid-line or while moving, the timer remains the only way out; catastrophic faults keep **R** = new trip.

v1.9.1 patch — announcement chaining The announcement queue has been hardened: every PA in the catastrophic chain (`tech_incident` → `dim_light` → `evac`) now waits for the previous one to *fully* finish before the next one fires, so messages are never cut off. The emergency brake squeal (`brake_noise`) no longer loops forever once the cabin is parked out-of-service. The state machine is now **five phases** with the new `dim_announced` phase between intervention and evacuation.

13 Auto-exploitation (X mode)

13.1 Concept

The `X` key hands driving over to the simulator: boarding, door-close chime, buzzer, trip, passing-loop crossing, arrival, turnaround, then next cycle — round after round, from the published opening time (08:45) to the last scheduled descent (16:45). Meant to be left running in the background: you hear the cabin ambient, announcements, buzzers and crossing whoosh exactly like a real day. On arrival the automaton waits for the **cable to settle** before opening the doors (rebound amplitude below 2 cm: 17 to 26 s at the lower station, where the rope is 3.4 km long; almost immediate at the top, except that the counterweight is then swinging at the bottom), then allows 12 s for alighting before the turnaround. The panel reads “SETTLING ±12 cm”.

13.2 Activate from any state

`X` can be pressed in any situation:

- **At a terminus** — a new boarding cycle starts.
- **Mid-trip in the tunnel** — takes over the current trip (TRANSIT phase); the regulator picks peak / off-peak cruise from the wall clock.
- **Stopped mid-tunnel after an incident** — pre-arms READY, fires the departure buzzer, and resumes automatically.

What still works. During automatic operation, driving is locked (setpoint, brakes, doors, direction), but the horn (`K`), headlights (`H`), lights (`J`, `C`), views, menus and `X` itself always respond — even when the keyboard focus stayed in the 3D window (after skier mode, for instance): it forwards all its keys to the PC.

13.3 24/7 mode (Shift+X)

`Shift+X` toggles the 24/7 override, which ignores published hours and runs the line continuously — useful for ambient background play outside the real ski-season window.

13.4 Side panel

An amber box appears bottom-right (next to the event log) whenever `X` mode is active. It shows:

- real-time clock, published hours, OPEN / CLOSED / 24/7 pill;
- current phase with live countdown (e.g. BOARDING 18 s, DOORS_OPENING 2.3 s);
- peak / off-peak band;

- daily counters: trips, pax, distance (km).

Border is green during service hours, orange out of hours, purple pill when 24/7 is armed.

13.5 Speed adapted to the clock

12 m/s during rush windows (9 h–12 h and 14 h–16 h), 10.3 m/s off-peak — reproducing the real 7 min 54 trip time.

13.6 Passenger load by time of day

Ascents are heavy around 10:15 in winter (15 to 95% of capacity depending on the hour, with a ±15% random spread). In winter, descents stay almost empty (0 to 10%: skiers ski back down). Only summer skiing on the glacier (June 15 to July 31) changes this: ascents heavy from opening (peak around 08:30) and *descents* heavy around noon, when the glacier closes. The effective mass feeds the physics: cable tension reflects a realistic day. The Web version applies the same law.

13.7 Input lockout

Cockpit buttons and most keys are masked during auto mode to avoid fighting the state machine. Only the following pass through: `X`, `P`, `N`, `L`, `Esc`, `F1`–`F9`, `F11`, `O`, `+`/`−` (zoom), `Backspace` (abort announcement).

13.8 Trip log (F5)

`F5` opens an EXPLOITATION LOG dialog with two tabs:

- **Trips** — last 100 round-trips: departure / arrival timestamps, direction, pax, cruise speed, distance, duration, incidents.
- **Daily stats** — last 60 days: trips, pax, kilometres.

Data is read directly from `exploitation.db` (SQLite, WAL, `TRUNCATE` on close).

14 The skier

In a nutshell

You leave the driver's seat to play a skier, skis in hand: they walk through the stations, climb the stairs, board the train, go to the driver's seat and exit at the top. Meanwhile the funicular runs by itself.

14.1 Starting and leaving

- **PC**: `F9` or the SKIER button, in the 3D view (`F4`) with a trip started. `F9` again, or QUITTER, to drive again.
- **Web version**: SKIEUR button (key `K`).
- CONDUIRE (offered next to the driver's seat of the driven train): you sit at the controls; the skier and auto-operation stop.

Back in skier mode, you find them where you left them, even in a train that has moved meanwhile. The first start is on the Val Claret square when going up, and on the top terrace, at a table in front of a plate of chips, when going down.

14.2 Controls

PC	Tablet
`Z` `Q` `S` `D` (or `W` `A` `S` `D`), arrows: walk	Joystick, bottom left
`Shift`: run	COURIR
`V`: 1st / 3rd person	1RE PERS.
`E`: skis on / off	CHAUSSER
Drag in the 3D, wheel: look, zoom	A finger dragged anywhere but on the joystick

On foot as on skis, the direction is read in the view of the moment you put your thumb down (or press a key); turning the view without letting go does not deflect the walk. On a tablet, only a finger placed away from the joystick turns the view; looking up is limited and the horizon comes back as you walk.

On PC, in skier mode, letters make them walk: only the function keys, `Esc`, `P`, `N`, `L`, `J` and `C` keep their role, and the dashboard buttons can still be clicked.

14.3 Where they can go

- **Val Claret**: the square, the stairs, the automatic entrance door, the waiting room and its sliding doors, open during boarding, the landing, the stepped platforms, the car; at the end of the platform, the pit under the track (the head goes below the rails) and its stairs.
- **In the train**: they walk in it, are carried along, including through the turnaround at a station; they can go to the driver's seat. The orange ski racks do not stop them: you walk through them (walking in the train was "a pain").
- **Grande Motte**: the platforms, the landing, the automatic bays of the head wall, the terrace and its staircase down to the snow, at the end of the point; at the bottom of the left platform (looking uphill), the Génépy piste door, its corridor and the snow level with the sill.

They climb steps of 40 cm at most. If they fall into a hole, they come back to the last safe ground.

14.4 The automatic modes, plainly

Three automatisms, three names (since 1.18.5):

- **Autopilot** (PC, key `A`, PILOTE pad): *one trip* by itself — doors, READY, 100 %, arrival, doors, passengers off — then it hands back.
- **Automatic operation** (key `X` everywhere, skier mode included; Web also `F3`; the EXPLOIT. button, on the dashboard and in the skier's bar): *the whole service* — boarding, timetable, departures one after another, the faults of Faults mode. Switching the skier on or off, or taking the driver's seat (CONDUIRE), never changes it: `X` stops it, and the setpoint then goes back to 100 %.
- **The skier's loop** (Web and the PC's 3D view, key `B`, BOUCLE button): *the skier* does the loop alone — station, train, terrace, ski, back — and switches automatic operation on for his ride. Started in the waiting room or on the platform, he stays there; while the platform door is shut he waits sitting on a bench, and runs to the train as soon as it opens. Started on the pistes, he puts his skis on and goes down. At the top he puts his skis on as soon as he reaches the snow and skis like an expert: carved turns, up to 100 km/h, along a route that follows the marked pistes down to the foot of the Val Claret station, where he takes them off

and runs to the gate. Leaving the skier view to change view does not stop it: he carries on backstage (watch him from the exterior view) and you find him where he is when you come back. CONDUIRE or BOUCLE again stop it.

Driving by hand is none of the three.

Inside the train, you walk from one car to the other through the gangway opening in the coupling partition (0.90 m × 2 m), floor and bellows included. The floor is stepped, one landing per panel, level in the stations like the platform steps.

14.5 The funicular meanwhile

Switching to the skier, or back, **does not change the operation**: automatic or not, it stays as it was. Without auto-operation the train does not move until you start it (EXPLOIT., or the console); a train at the platform with its doors closed, outside a trip, opens them so you can board. Under auto-operation (on PC, 24/7):

- it **waits** for the skier as long as they are in the station outside the car, even standing in a doorway — 45 s at most, then it closes and leaves;
- as soon as they are **inside the car**, past the door line, it starts the departure sequence 1.5 s later: announcement, doors closing, buzzer, departure (about thirty seconds in all);
- if they were already on board on arrival, the stop lasts as usual: there is time to get off;
- at the top the doors open 3 s after the stop (the cable span is short there, the train does not bounce); at the bottom, once the cable has settled; as soon as they are open at the platform, the waiting room's sliding doors open too.

On PC, in skier mode, `O` and the VUE button are refused: the camera is the skier's, changing the view would only move the sound. The skier's EXPLOIT. button starts or stops auto-operation (on PC the dashboard's AUTO button does the same). Enabling auto-operation always returns to NORMAL mode. On PC, while the skier's scenery is being prepared (a few seconds the first time), the SKIEUR button shows the progress.

In the stations, background music: the orchestral overture while waiting at Val Claret, the Toreador song at Grande Motte (files in `sons/musique/`, outside the repository; on PC, when missing, they are fetched once from the Web version's server into the profile folder, `musique/`, and played locally from then on).

14.6 What they hear

In the train, the cabin sound. On the platforms and in the waiting room of **Val Claret**, once the train has left, no more cabin sound: regular gusts of air, whistling like a wire, followed by silences, due to pressure waves in the tunnel. On the platforms of **Grande Motte**, the machine room, exactly as in the machine-room view, but fainter the farther you are from the machinery (−6 dB per doubling of the distance beyond 4 m, floor −24 dB). Outdoors, the wind. On foot in the tunnel, the gusts.

The **departure buzzers** are platform loudspeakers: each station's buzzer sounds in its own station at every departure, and you hear it if you are there — platforms, waiting room, machine-room view, train at the platform — even when the train restarts from mid-tunnel; you never hear it from a train in the tunnel (inside or outside view), nor outdoors.

14.7 The tunnel, on foot

From a train stopped in the tunnel with its doors open (Challenge mode, a fault), step down from the door onto the service stairs, on the right looking uphill; at the chamber of roller 145 the wall is pierced: the circular **emergency gallery** leaves the tunnel at 90° and climbs straight (36 %

ramp) to the portal some fifteen metres away, where its floor meets the slope (skis on, down you go). tunnel, ÉVACUER (key `U`) removes the face's **emergency exits** — the two yellow D panels with a black rim either side of the windshield — : through the hole, down onto the track, and the service stairs lead back to a station or to the gallery. While someone is on foot in the tunnel or the gallery the train is **immobilised**, without limit: automatic operation waits, any departure is refused ("Skier on the track"); the panels are back at the next stop at a platform. Fallen into the pit between the rails, one climbs onto the slab and the stairs. A moving train reaching a pedestrian hits them: back to the station. They climb steps of up to 40 cm, or ledges of up to 1.25 m on foot outside the stations (out of the pit, over the rail); aboard, neither wheels nor bogies are visible, and the floor is sealed up to the closed doors.

14.8 The whole loop by itself

AUTO (key `X`): the skier does the entire loop alone — from the Val Claret square to the hall, the train (waiting for it at the platform), the ride, out at the top through the Génépy door, on foot to the start of track no. 4, skis on, down the track, skis off at Val Claret, back to the station, and again. AUTO again to take over.

14.9 Skiing

- **Skis on**: outdoors, on the snow, key `E` or the CHAUSSER button; refused in a station or a train. DÉCHAUSSER takes them off.
- **Winter**: in skier mode the terrain is under snow (except rock on steep slopes and valley floors below 1 800 m), shaded by the sun; the OpenStreetMap pistes are groomed and **marked**: a pole on each side every 40 m, in the piste's colour (green, blue, red, black). At the top of the screen: speed and the piste under the skis.
- **Gliding**: `Q`/`D` to turn, `Z` to push on the flat, `S` to snowplough. With the joystick (PWA), as in games: *point* where you want to go relative to the view *at the moment you put your thumb down* (up = straight on); that frame stays frozen until you let go, even while the camera swings behind the skis — a steady aim gives a steady heading; pulled towards you = snowplough; at a standstill, up = push; `Shift` or SCHUSS to tuck. The camera comes round behind the skis gently (50 °/s at most), never in the 2 s after you turned the view by hand, and does not move when you stand still. The skier takes the posture: crouched in a tuck, legs apart and skis in a V in the snowplough.
- **No falls**: you stop against a wall, that is all (falls were tried then removed: far too frequent on steep pistes). If you ended up under the snow on foot, CHAUSSER brings you back to the surface.
- **Signs**: on top of every third edge pole, on the right going down, a round sign like the real ones: a disc in the piste's colour with a white rim, the piste name in an arc at the top, TIGNES in an arc at the bottom, the marker number in the middle, *decreasing* towards the valley (1 at the bottom).
- **The piste map** (CARTE button, skis on): a semi-transparent overlay, north up, centred on the skier (red arrow in his direction): the pistes in their colour with their name, both funicular stations and its real route (underground, black-and-yellow dashes), both trains live, a scale bar. Zoom with the +/− buttons, the wheel, a pinch, or the +/− keys (`=` too), from 200 m to 17 km across.
- **Physics**: gravity along the slope; the edges hold what goes sideways to the skis; groomed-snow friction and air drag; the faster you go, the wider the turn. Station walls stop the skier.
- **The ghost**: five real descents from the glacier station to Val Claret (GPS tracks from April 2026, 4 min 45 to 10 min 15). A translucent blue skier starts when you set off from its start point, at the foot of the station; on arrival, near Val Claret station, both times are shown.

- **The loop**: at the bottom, take the skis off, walk to the station and ride the funicular back up.

14.10 Ski lifts

Since version 1.27, **48 lifts** of the ski area stand where they really are (all chairlifts, gondolas, the 3S and both aerial tramways; T-bars will follow). Among them: the **Marmottes Express** 6-seat chairlift (Bartholet, 11 towers, Val d'Isère), the **Boisses** 10-seat gondola (CWA Omega IV cabins, 11 towers, Tignes 1800) and, since 1.24, the **Olympique 3S** (Doppelmayr 2002, 30-seat CWA Zeta cabins, two track ropes per side and a haul rope, 3 lattice towers, 1835 → 2688 m, Val d'Isère) and, since 1.25, the **Grande Motte aerial tramway** (Von Roll 1975, two 115-passenger cabins shuttling and crossing, 10 m/s, 5.2 m/s over the tower, 3034 → 3456 m). Walk in; you ride standing. They show up in skier view.

- **Towers**: the conical mast, the ladder and the catwalks are modelled. Each tower head carries its real number of sheaves on the uphill and downhill sides: *support* (the rope runs over them), *compression* (under them) or both.
- **Haul rope**: it forms a loop and sags between towers. Tower heights are not published anywhere; they are computed so that the rope stays at least 6.5 m above the ground, a little less near the stations where it comes down to platform height. At each station entrance a sheave battery bends it in an arc from the level station to the slope of the line.
- **In the stations**, each carrier *detaches*: the cam opens the grip and it rolls on the rail. Tyres slow it to 1 m/s for loading, it goes round the bullwheel, then it is accelerated again and the grip closes on the rope. The safety bar is automatic: it rises at the end of braking and comes down at the start of the departure accelerator. Gondola doors open while loading.

Boarding: the gondola is boarded on foot. Walk in while the doors are open: if you have skis on, you take them off and they go into the rack on the door. You are seated on the bench facing uphill; at the top you stand up when the doors open and walk out. The chairlift is boarded with skis on (they are put on at the gate if needed). Skis cannot be put on or taken off while riding. For the chairlift, wait behind the yellow gates. When they open, move up to the stop line: the chair comes from behind and seats you by itself; with skis on, they rest on the footrest. At the top you stand up at your seat and glide down the short unloading slope, out of the chair's way. In skier view, speed and altitude are always shown, with the lift's name while riding it.

Control desk: near a lift or on board, the POSTE button opens its desk: rope speed, power, service and bullwheel brake status, speed setting (low, medium, high — high by default), DÉMARRER (start), ARRÊT (service stop, about 14 s), ARRÊT D'URGENCE (emergency stop on the bullwheel brake, a few seconds, chairs swing) and RÉARMER (reset, once stopped). The desk follows the nearest lift.

Getting there: open the map (CARTE button, on foot or on skis) and zoom out with "" or the wheel. Lifts are drawn in black with their bottom station in red. Tap or click one to be dropped at the foot of its bottom station.

The models for each manufacturer are ready: Doppelmayr, Poma, Garaventa, Bartholet and Leitner chairs, bubble chairs, cabins, T-bars and platter lifts, 3S and aerial tramway. The other lines will follow.

14.11 Coming next

The driver's seat of the other train, the ski lifts, jumps and falls. Tracked in `SKIEUR.md`.

15 Mid-tunnel abnormal-stop sequence

Any latched stop engaged *while rolling in the tunnel* — manual electric stop (`3`), emergency stop (`4`), overspeed auto-trip, dead-man vigilance loss, or a service-stopping fault (door, thermal, motor degraded, aux power, parking drum stuck, fire) — now follows the full real-world Von Roll safety-chain protocol:

1. READY / ghost-ready / departure buzzer cleared immediately;
2. cabin decelerates under its own physics (coast for electric stop, rail brakes for emergency, motor cut for faults);
3. once $|v| < 0.1$ m/s, the **incident announcement** plays — not when the button is pressed, but when the cabin has actually stopped (`tech_incident` in general, `dim_light` + `evac` for fire);
4. `trip_started` flips back to FALSE: the trip is formally suspended;
5. the **parking drum** engages — raising the speed setpoint no longer moves the cabin on its own;
6. the driver clears the cause (stop, fault), presses READY (triggers the "Remise en route" announcement), then DEPART (buzzer, drum releases, trip resumes).

At the termini the sequence short-circuits: messages play instantly and the drum is not required, as the doors-open parking state already holds the cabin still.

Silent advisory faults (cable tension, wet rail, slack cable) remain dashboard-only — no PA, no forced stop.

16 Auto-update

16.1 Principle

At startup, the program silently queries the releases page on GitHub. If a newer version exists, it offers:

"Version X.Y.Z is available. Install now? [Install] [Later]"

It then downloads the update archive for *your* system (the program and the 3D view, a few tens of megabytes), puts it in place itself and restarts. The mechanism is the same on Windows, macOS and Linux; the release notes are shown in the interface language.

You may also check anytime via **Help → Check for updates**.

16.2 Your data

Operations log, best scores, settings, chosen language and the consent to send reports live in your profile (`%APPDATA%\PerceNeigeSimulator` on Windows, `~/Library/Application Support/PerceNeigeSimula` on macOS, `~/.local/share/PerceNeigeSimulator` on Linux): neither an update nor a reinstallation touches them. What versions up to 1.19 kept in the program folder is moved there at first start.

16.3 Technical safety

The archive only comes from GitHub (address checked), over verified HTTPS; paths leaving the folder are refused; the previous version is set aside and restored if anything fails.

17 Anonymous bug reports

17.1 Automatic capture

On program crash, a JSON report is saved into `crash_reports/`. Everything is anonymised before send: paths become `~`, username stripped, no IP, no email.

17.2 Sending

Nothing is sent without your consent. The first time, the program asks you; if you accept, the anonymised reports (crash, freeze, failed start) are sent by themselves. You can change your mind at any time with **Help** → **Send incident reports automatically**.

Manual report via **Help** → **Report a problem**: description and reproduction steps.

18 FAQ

Train does not start

Check: doors closed (`D`), electric stop disarmed (`3`), emergency stop disarmed (`4`), command > 0 (up arrow).

I arrived, I want to return

After arrival, press `R` for a new trip in the opposite direction.

No sound

Key `N` to mute / unmute. On Linux, install `qt6-multimedia` and a GStreamer plugin.

The 3D view (F4) opens in a separate window

On Linux, embedding the 3D view inside the main window is only possible under X11: Wayland offers no equivalent of the XEmbed mechanism, and a native Wayland Godot window does not even expose a usable X identifier (XID). The simulator therefore switches to XWayland automatically at startup and launches the Godot viewer with `--display-driver x11`. If the 3D view still appears detached, check that `xdotool` is installed and that the `PERCE_NEIGE_KEEP_WAYLAND` variable is not set (it forces the native Wayland backend, hence a separate 3D window). On Windows, embedding goes through `SetParent` and is unaffected.

Change language

Key `L` toggles FR ↔ EN. System locale is auto-detected at startup.

How do I become the skier?

On PC: switch to the 3D view (`F4`) with a trip started, then `F9` or the SKIER button. In the Web version: the SKIEUR button. See section 14.

The 3D console buttons do nothing (PC)

Fixed in v1.16.3: the 3D view now passes every press to the simulator. In auto-operation (`X`), the train stays with the automaton, as with the keyboard.

The window does not fit the screen, or it is too small

The interface adapts to the detected screen. To force it: **Display** → **Interface size**; `F11` for full screen.

The 3D stutters, or looks blurry

Quality adjusts itself (the log says which level it is at); you can force it lower — or higher, if it is too blurry — in **Display** → **3D quality**. On PC the rendering never drops below 85 %.

My comfort score is bad

You accelerate or brake too brutally. The score is based on the integral of *jerk* (acceleration derivative). Tip: let the regulator handle things, do not touch the brake on approach, raise the command in steps.

Licence / License

MIT. ARP273-ROSE — 2006 TI-Basic original, 2026 PyQt6 port.

Sources

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